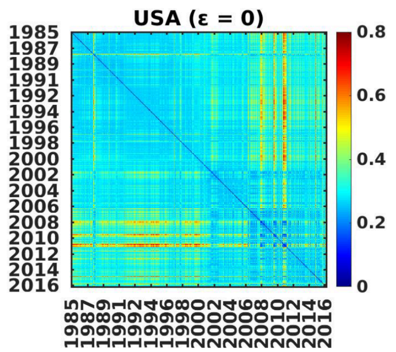
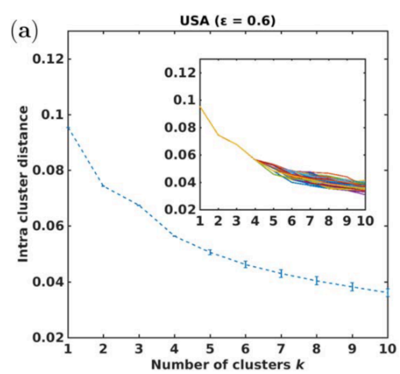
Assignment-1
(Marks=10)
Data Science in Financial Markets

1. Download stock data from the S&P-500 index for 10 years based on the starting date mentioned in the DSFM groups:
   https://docs.google.com/spreadsheets/d/1Kypnnxs81Bu0Gh0eq5PxmFgJydl-FYetJyB4iq2iFXo/edit?usp=sharing
2. Construct one CSV file of all the stocks for close prices (similar to the attachment SP500_350.csv) and arrange them in sectors.
3. Plot all the closing time series in 3X3 subplots, i.e., 9 time series each, as discussed in class.
4. Remove all the stocks with more than two consecutive days' NAN values.
5. Plot all the log return-time series *[r(t)=lnP(t)-lnP(t-1)]* in 3X3 subplots as discussed in class.
6. Remove all the stocks with |r(t)| > 0.8.
7. Construct a correlation matrix for the full-time horizon of 10 years using log returns.
8. Construct correlation matrices using r(t) of epoch size 20 working days and with a shift of 10 days.
9. Construct a similarity matrix S(t1,t2)= <| C(t1) - C(t2) |>.
10. Construct a 3D Multidimensional scaling plot using the Similarity matrix "S".

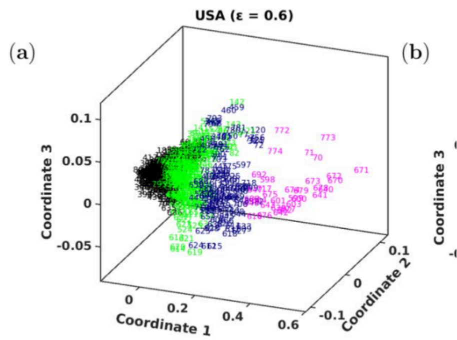


11. Find an optimum number of clusters using 1000 different conditions for the elbow method of k means clustering. eg.

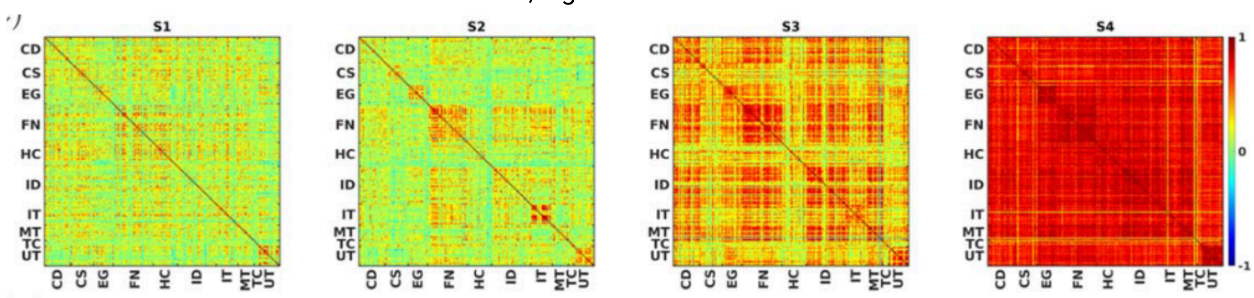


12.

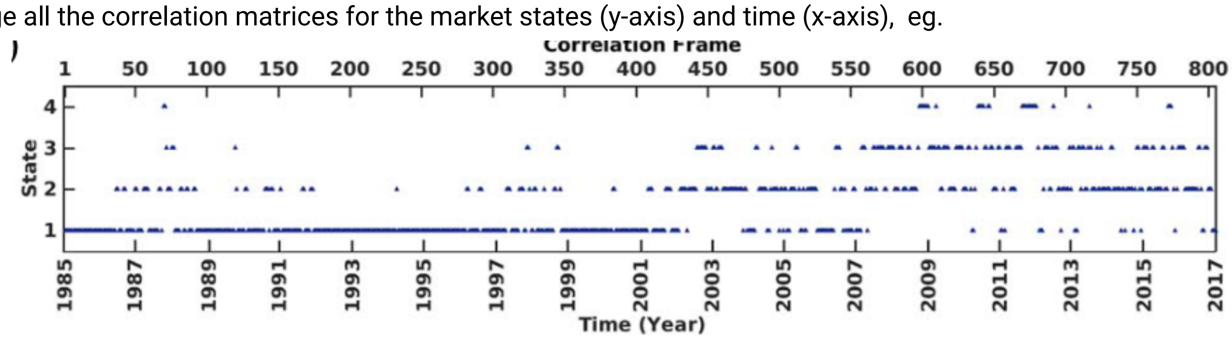
13. Plot k-mean clustering results with optimum value of clustering found in No. 10 (above) eg.



14. Plot typical correlation structures of different states, eg.



15. Arrange all the correlation matrices for the market states (y-axis) and time (x-axis), eg.



16. Show the transition of consecutive market states, eg.

	S1	S2	S3	S4
S1	365	47	7	1
S2	51	144	35	1
S3	4	40	69	7
S4	0	0	9	24

17. Upload the Python code with the assignment and mention the total number of lines of the code.

Note: All x-labels and y-labels should be matched with the given template figures as examples for a few questions.